BINHUA HUANG

Curriculum Vitae

Center for Intelligent Bionic
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My Webpage
Github

Education

April,2022 **M.S. Computer Science**, *University of Science Malaysia*, GPA: 3.47/4.
Courses: Research and Empirical Methods in Computer Science, Advanced Algorithms & Complexity, Machine Learning, Wireless and Mobile Communications, Internet of Things
Dissertation: Tactile Sensing Using Machine Learning Methods (109 pages, in English)

July,2015 **B.E. Electronic Information Engineering**, *Longyan University*, GPA: 79.33/100, top15%.
Courses: C Language, Object Oriented Programming, MATLAB Programming, Embedded Systems and Applications, VHDL Language, Signals and Systems, Electronics, Digital Signal Processing, Automatic Control, Communication Principle, Sensors, Microcomputer etc.
Thesis: An IoT System Based on Cloud Computing (Best thesis award, 2%)

Publications

Published:

2022 **Modeling and control strategy of flexible joint servo system in humanoid manipulator driven by tendon-sheath**.
Meng Yin, Dongyang Shang, **Binhua Huang**, Xinyu Wu, and Wei Fen

2022 **RBF network-based adaptive sliding mode control strategy for the tendon-sheath driven joint of a prosthetic hand**.
Meng Yin, **Binhua Huang**, Zhengkun Yi, and Shibo Cai

Accepted:

2022 **Depth Recognition of Hard Inclusions in Tissue Phantoms for Robotic Palpation**.
Zhenning Zhou, Senlin Fang, Chaoxiang Ye, Tingting Mi, **Binhua Huang**, Xiaoyu Li, Zhengkun Yi, and Xinyu Wu
The manuscript was accepted by *IEEE International Conference on Real-time Computing and Robotics (RCAR 2022)* and as the best student paper.

2022 ***Multi-kernel-size Convolutional Supervised Autoencoders for Tactile Gesture Recognition***.
Chaoxiang Ye, Xiaoyu Li, **Binhua Huang**, Yuanzhe Su, Tingting Mi, Zhenning Zhou, Zhengkun Yi, and Xinyu Wu
The manuscript was accepted by the editor of *IEEE International Conference on CYBER Technology in Automation, Control, and Intelligent Systems (CYBER 2022)*.

In revision:

2022 ***Methods to Recognize Depth of Hard Inclusions in Soft Tissue Using Ordinal Classification for Robotic Palpation***.
Zhenning Zhou, **Binhua Huang**, Runzhi Zhang, Meng Yin, Chengliang Liu, Yiwen Liu, Zhengkun Yi, and Xinyu Wu
The associate editor of *IEEE Transactions on Instrumentation & Measurement* has asked for major revisions, and now we are revised and submitted the responses.

Under review:

2022 ***Liquid Volume Estimation by Tactile Sensor Using Machine Learning***.
Binhua Huang, Meng Yin, Zhengkun Yi, Chaoxiang Ye, Xiaoyu Li, Zhenning Zhou, and Xinyu Wu
The classification accuracy for the task of liquid volume estimation reached 99.78% with CNN; it is a considerable improvement compared to the state-of-the-art vision-based solution. The paper was subbmited to *IEEE Transactions on Instrumentation and Measurement*.

2022 ***An IoT Based SCD Risk Rescue Alert System***.
Binhua Huang, Selvakumar Manickam, and Shafiq Ul Rehman
We proposed an IoT system that can detect the Sudden Cardiac Death (SCD) in real-time and ask for help automatically. The paper was subbmited to *KSII Transactions on Internet and Information Systems*.

Manuscript in preparation:

2022 ***An Optical Wearable Device For ASL Recognition Using Neural Network***.
Binhua Huang, Chaoxiang Ye, and Zhengkun Yi
We designed a novel wearable device for American sign language (ASL) recognition, reaching 99.9% accuracy. It has the potential to be used in real-time ASL translation for deafmultism films or TV shows. The experiment was completed. Now in the writing process.

2022 ***A Novel Tactile Sensor Mimics the Function of Ruffini Receptors Applied in Localization***.
Binhua Huang, Chaoxiang Ye, Meng Yin, Zhengkun Yi, and Xinyu Wu
We propose a new tactile sensor and use a neural network to localize touch areas and forces.

Research Experience

Shenzhen Institute of Advanced Technology, Chinese Academic of Science

August 2021 - Present ***Visiting Student (as well as RA)***.
Developing a deep multi-modal architecture for accurately predicting protein interaction information from biomedical text.

Advisor : **Dr. Zhengkun Yi**, *Professor, Center for Intelligent Bionic*, SIAT (*Personal Web-page*)

Skills

Programming Languages	Python, C, C++, VB
Frameworks and Platforms	Sci-kit, Pytorch, keras, Linux, Universal Robotics, MCUs, Arduino, Raspberry Pi, FPGA
Software and Tools	Matlab, LaTex, Altium Designer, Multisim, Proteus, Quartus II, Keil, IAR
Natural Languages	Chinese (mother tongue), English (CET6, M.S. degree conducted in English)

Fellowships & Awards

2020	Second prize of Fujian provincial innovation competition
2015	Best graduation thesis award
2014	Third-class scholarship
2012	First prize of campus electronic competition
2012	Third-class scholarship
2011	Third-class scholarship

Referees

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